

Universal Model of Urban Street Networks

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Analyzing 9000 urban areas' street networks, we identify properties, including extreme betweenness centrality heterogeneity, that typical spatial network models fail to explain. Accordingly we propose a universal, parsimonious, generative model based on a two-step mechanism that begins with a spanning tree as a backbone then iteratively adds edges to match empirical degree distributions. Controlled by a single parameter representing lattice-equivalent node density, it accurately reproduces key universal properties to bridge the gap between empirical observations and generative models.

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Introduction—Urban planners, engineers, and geographers measure and model street networks to describe or reproduce the structural characteristics that underlie urban transport, accessibility, and sustainability. These models usually incorporate geometric and topological information, and may be empirical or generative. The former usually use governmental data or open data to describe real-world networks. Various properties of these networks were studied over the past two decades [1–22], and the temporal evolution of these structures is also studied through the digitization of historical records [10,11,23–27].

Despite this empirical research, a gap remains in developing parsimonious models that can replicate the heterogeneity of observed urban street network properties. Such models are crucial for exploring the impact of urban form on various processes such as CO₂ emissions, traffic congestion, and the macroscopic fundamental diagram. Previous authors proposed simple models based on local optimization processes and the mechanics of leaf pattern formation [8,28,29], extended in [30]. Recent artificial intelligence (AI) research has synthesized street network images and graphs with deep generative models [31–33]. Such methods usually test the visual and statistical similarity of certain street network properties such as city block area, compactness, and aspect ratio [31]. While useful, these models require extensive input data for training and potentially fail to capture unique properties that distinguish street networks from general spatial networks [31,34–36].

Can a simple spatial network model capture the most essential features of real-world street networks? To address

this, we must first identify which features are truly essential. The primary function of street networks is to facilitate smooth traffic flow between locations, a process often governed by shortest paths. Accordingly, a key property of street networks is the distribution of shortest paths (weighted by distance), which can be characterized by betweenness centrality (BC). We propose that traffic dynamics are primarily driven by network connectivity, best described by BC-related measures. Therefore, a generative model of street networks should prioritize reproducing the BC properties observed in real-world networks.

Characterizing street networks—We thus first identify and measure key empirical street network properties. As discussed in [37], many measures commonly used for complex networks are not applicable to spatial networks in general, and street networks in particular, due to physical constraints [9,38]. Street networks are characterized by a narrow degree distribution and a high clustering coefficient [39]. Traditional indicators from transportation geography [1] tend to be redundant, as they largely reflect variations in the average degree. A more informative characterization of street networks relies on geometric and structural measures, such as the distribution of street segment lengths, block shapes, and street angles [39]. The most fundamental quantity, the node count N , has been shown to grow sublinearly with population size [10,11], with a 1% increase in urban population typically resulting in a 0.95% increase in the number of intersections [22]. Furthermore, the total street length scales as $L \sim \sqrt{A_u N}$, where A_u denotes the urbanized area [8,40].

Degree distribution and betweenness centrality—A basic characteristic of street networks is the fraction f_i of nodes with degree $i = 1, 2, 3, 4$. Specifically, f_1 represents the

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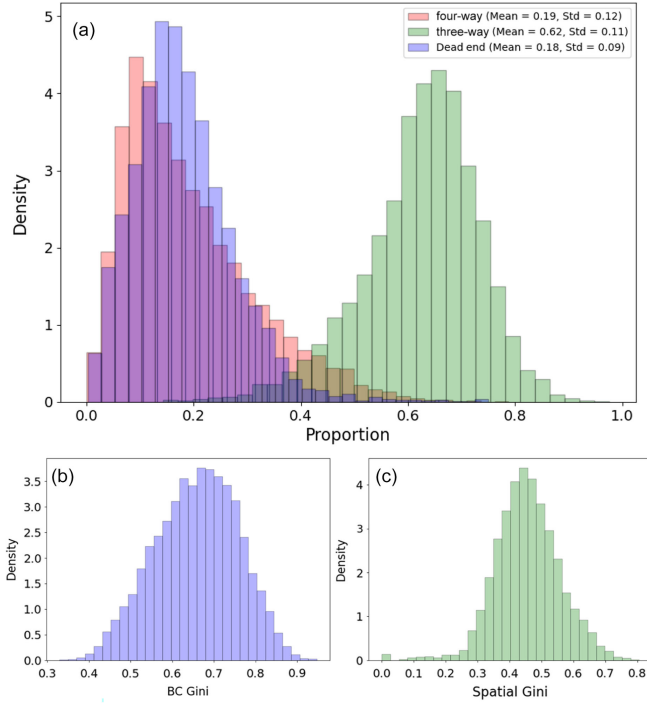


FIG. 1. Empirical probability distributions of the fractions f_i of nodes having a degree $i = 1, 3, 4$ (f_2 not shown) for ~ 9000 urban areas worldwide. The average values are $\bar{f}_1 = 0.18$, $\bar{f}_2 \approx 0.0$, $\bar{f}_3 = 0.62$, $\bar{f}_4 = 0.19$. (b) Probability distribution of the Gini coefficient for the BC and in (c) for the spatial Gini coefficient.

proportion of dead ends, while f_3 and f_4 correspond to three-way and four-way intersections, respectively. The fraction f_2 of degree-2 nodes is irrelevant as these have no real impact on the graph's topology (in practical applications, these nodes are sometimes kept for describing a change to street attributes in the middle of a graph edge, such as a change in lane counts, directionality restrictions, etc., which is why f_2 can be slightly different from zero).

An effective model should accurately reproduce the statistics of this degree distribution. We first study the empirical properties of approximately 9000 urban areas worldwide (this dataset comes from [22]; see also Refs. [41] for more details) and show that these quantities fluctuate slightly around their means (Fig. 1). The degree distribution, however, does not fix the topology and many different networks are possible. An important characterization of flows on networks is given by the BC, defined for edges as [42]

$$g(e) = \frac{1}{N} \sum_{s \neq t} \frac{\sigma_{s,t}(e)}{\sigma_{s,t}}, \quad (1)$$

where $\sigma_{s,t}$ denotes the number of shortest paths from node s to node t , and $\sigma_{s,t}(e)$ represents the number of those paths that pass through edge e .

We use shortest geometric paths that minimize total distance. Alternative criteria—such as travel time—could

be considered by assigning velocities to edges. When velocities are constant, minimizing distance is equivalent to minimizing travel time. Differences arise only when velocities vary across links, as is typical in motorized transport with traffic. However, in many cases (e.g., walking), travel speeds are approximately uniform. Incorporating speed limits would constrain the model to vehicular transport and introduce additional parameters, reducing its generality and parsimony. For these reasons, we leave such extensions for future work.

Edges with high BC represent those traversed by many shortest paths, effectively functioning as main roads within the network. This quantity has been extensively studied [11,14,39,43], and a key finding is that street networks can often be decomposed into a “backbone”—typically represented by a minimum spanning tree (MST)—characterized by high BC, while the remainder of the network exhibits significantly lower BC. Edge BC correlates with age [11], suggesting that street networks tend to evolve from a backbone of older streets, and that over time, additional edges connect to this backbone, increasing the network's density. This observation highlights that BC is highly heterogeneous in most street networks, and any reasonable model for these systems should, therefore, capture this specific property. However, we do not require a model to reproduce a specific BC probability distribution, but rather to reflect its heterogeneity. To quantify the heterogeneity in the BC distribution, we calculate the Gini coefficient G , which ranges from $[0, 1]$ (see Ref. [41]). A value of $G = 0$ corresponds to a uniform BC across all edges, whereas $G = 1$ represents extreme heterogeneity, where a single edge carries high BC while all others have values close to zero. We compute these values for the street network of each urban area in the world, using the street network data from [22]. Our empirical analysis reveals that for most cities worldwide, G is notably large: these networks consistently feature a small number of bottlenecks. Specifically, we find a median G of approximately 0.66, indicating a pronounced heterogeneity [see Fig. 1(b)]. In addition to G , we compute the spatial Gini coefficient G_{spa} , which provides a measure of spatial heterogeneity, by dividing the study area into a regular grid. For each grid cell c , we determine the number of network nodes N_c contained within it [the value of the spatial Gini is only weakly affected the grid size; see Supplemental Material (SM) for details [41]]. The spatial Gini coefficient is then calculated from the distribution of these N_c values. G_{spa} values are also large, with a median around 0.5 [Fig. 1(c)].

Equivalent lattice density—To construct a simple street network model, we map a street network to a grid. Lattice-based networks offer a simplified but powerful representation of real-world systems. If we want to model an equivalent street on a regular grid size, we need to estimate the probability that a node of the lattice is present. This equivalent lattice density p is estimated in the following

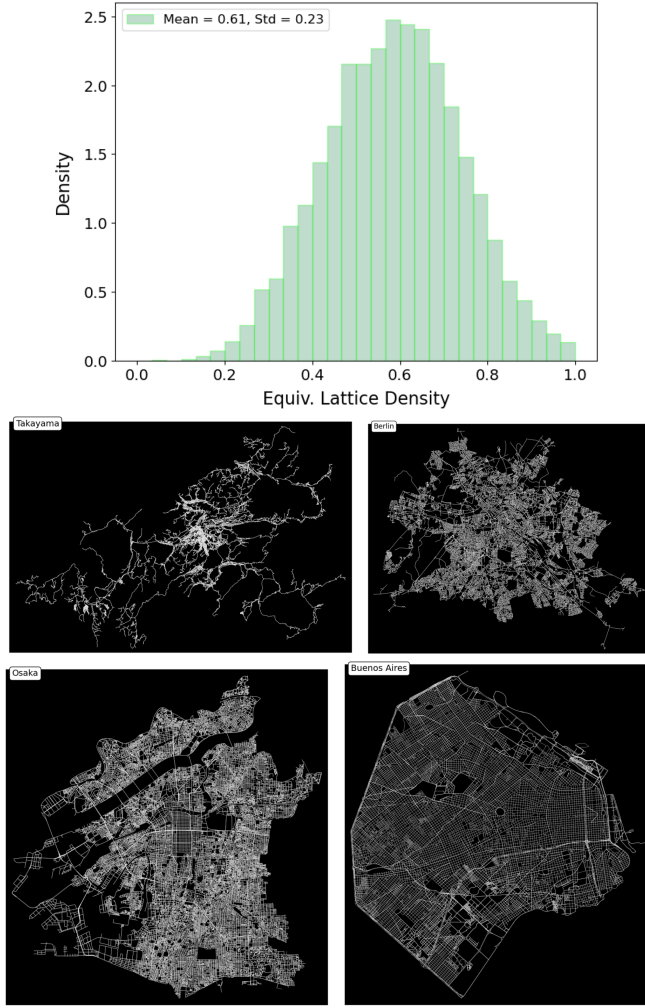


FIG. 2. Top: equivalent density distribution obtained for about 9000 worldwide urban areas. Bottom: examples of cities whose street networks exhibit various values of the parameter p : Takayama ($p = 0.22$), Berlin ($p = 0.52$), Osaka ($p = 0.69$), Buenos Aires ($p = 0.95$). For the method used to compute the value of p see the illustration in [41].

way. The number of nodes of a real-world network N_{emp} represents the intersections or end points in the street network under study (of the city, the urban area, or any type of area). We denote by ℓ_1 the average edge length, and the convex hull area A which is the smallest convex polygon that encloses all the nodes in the network (see SM for more details [41]). If the network were on a regular square lattice, the total length of the square lattice would then be $\sqrt{A}$ and the grid length would be ℓ_1 . The maximum number of nodes would then be $N_{\text{max}} = (\sqrt{A}/\ell_1)^2 = A/\ell_1^2$. The equivalent lattice density is then defined as

$$p = \frac{N_{\text{emp}}}{N_{\text{max}}} = \frac{N_{\text{emp}} \ell_1^2}{A}. \quad (2)$$

This quantity p is dimensionless and combines the density of nodes and the average street length. It is $0 \leq p \leq 1$: for $p = 1$, the network is equivalent to a regular lattice, while for smaller p , it is made of smaller clusters that are locally regular. This parameter provides a standardized way to compare street networks across different cities by accounting for the number of nodes, the average street length, and the area they cover. Figure 2 shows the empirical distribution of p around the world, alongside examples of cities with various values of p (other examples can be found in SM [41]). The average value is $\bar{p} = 0.61$ and the standard deviation $\sigma = 0.23$.

Universal model—A realistic street network model must match the degree distribution and reproduce key traffic characteristics across values of p , notably by capturing the strong heterogeneity observed in BC. This requirement imposes a nontrivial constraint: while it is easy to construct a spatial grid network that satisfies node-level constraints f_i , incorporating BC heterogeneity is more difficult. A nearly regular lattice, for example, produces a BC that decays smoothly from the center, yielding low heterogeneity and a small Gini coefficient—far from what is seen in real networks. In order to illustrate this, we used the random Eden model, a simple baseline choice for modeling a road network. This model generates a single connected component through local growth. After the growth phase,

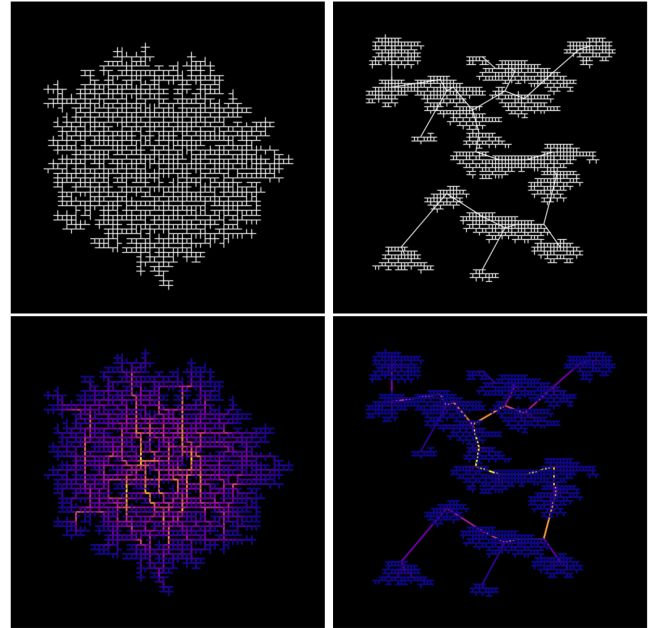


FIG. 3. Two different networks with both having an equivalent density $p = 0.5$ ($N = 4000$). On the left, we show a random Eden-like model, and below, the corresponding distribution of the Gini coefficient. In this case the Gini coefficient of the BC is $G = 0.51$, and the spatial Gini is $G_{\text{spa}} = 0.26$. On the right, we show the result for our model based on the MST and the corresponding BC map. The BC Gini is here equal to $G = 0.72$, and the spatial Gini $G_{\text{spa}} = 0.40$.

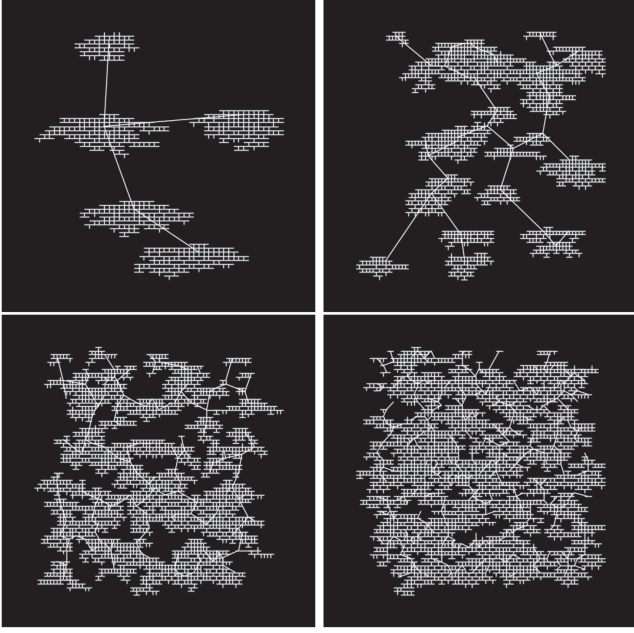


FIG. 4. Examples of networks generated by the model for $p = 0.23, 0.45, 0.76, 0.92$ and $N = 4000$. Target values for the fractions are the empirical ones: $\bar{f}_1 = 0.18$, $\bar{f}_2 = 0.0$, $\bar{f}_3 = 0.62$, and $\bar{f}_4 = 0.21$.

the degree proportions f_i can be adjusted to match empirical values (see SM for details [41]). While other options exist—such as percolating clusters or random geometric graphs—few produce structures with spatial and topological features comparable to real street networks. We show in Fig. 3 a network constructed as a random Eden model [44]. In this case, BC primarily concentrates at the network’s center and gradually decreases with distance, resembling the behavior of a regular 2D lattice. Additionally, the heterogeneity remains moderate, with a Gini coefficient of approximately $G \approx 0.5$. As the cluster is compact, the spatial Gini is also low with a value $G_{\text{spa}} \approx 0.26$ here. Taking a lattice and removing at random edges (such as in percolation) would lead to a small value of G for the same reasons.

This demonstrates that any compact cluster structure fails to produce sufficiently large BC heterogeneity. To achieve the observed heterogeneity, we follow insights from [14,37,39,43]. Our approach begins by constructing a high-BC backbone over a limited subset of nodes, and then incrementally adds edges to match the empirical node degree fractions f_i . More precisely, our algorithm (see pseudocode in SM [41]) starts by randomly selecting a set of grid points and generating a minimum spanning tree over this set based on Euclidean distances—motivated by results in [43], which highlight the role of MSTs in generating high-BC edges. The network is then iteratively modified through edge additions and deletions to achieve the target degree proportions f_i , while ensuring that the total number of nodes converges to a value consistent with the specified

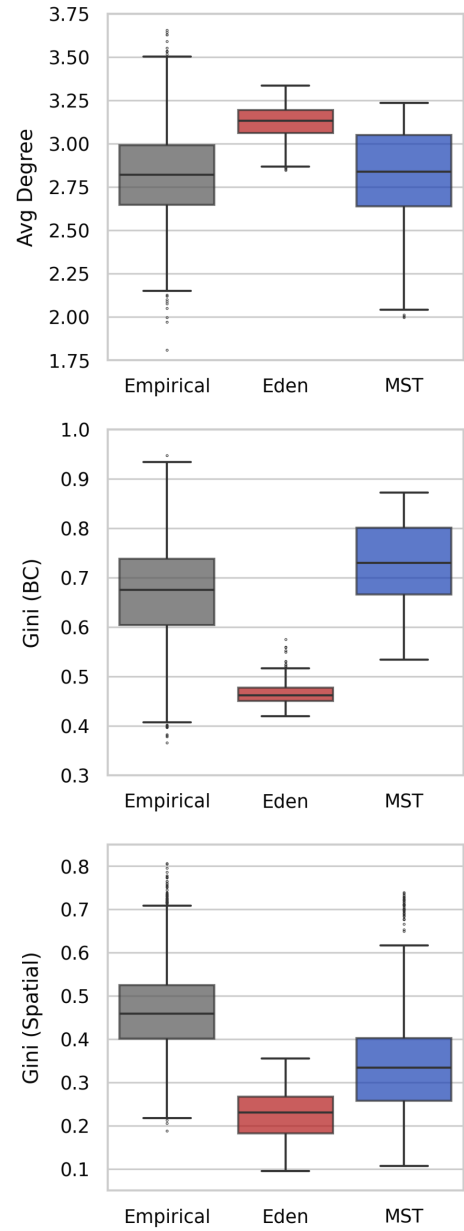


FIG. 5. Box plots for the average degree, BC Gini, and spatial Gini values from the empirical data, Eden model, and MST-based model ($N = 2000$ and 50 configurations for the models). Compared to the Eden model (as a baseline) across a full range of p , the MST-based model better reproduces the empirical distributions of these properties’ values.

grid density p , within a given tolerance. To incorporate empirical variability, we define $f_i = \bar{f}_i + \epsilon_i$, where $\bar{f}_i$ is the empirical average and ϵ_i is a small noise term satisfying $\sum_i \epsilon_i = 0$.

Figure 4 shows examples obtained for different densities p computed with this model. For low densities, the structure exhibits a predominantly modular organization, consisting of small clusters connected by “primary roads” derived from the MST. This configuration sustains a high BC. As density increases, clusters grow and gradually

merge. However, the resulting structure does not form a single compact cluster, as seen in models like the Eden growth process, for example. Instead, the merged clusters remain interconnected through narrow links, which continue to support a high BC. We observe that these distinct structures can be identified in real-world cities, as illustrated in various cases shown in SM [41].

Once the model converges, we measure various properties—including the average degree, G , and G_{spa} —and compare them with empirical data. Figure 5 shows that the MST-based model reproduces the empirical properties, providing a good approximation of their distributions, but somewhat underpredicts the spatial Gini coefficient (our simple model uses a regular lattice, whereas real networks exhibit greater spatial flexibility). We also checked that breaking the degeneracy of shortest paths on the lattice does not modify our results (see SM [41]). Importantly, however, the MST-based model consistently outperforms the Eden model (as a baseline) at reproducing each of these properties. In [41], we also present bivariate relationships of each property as a function of p , but these are not the focus of the proposed model—particularly given the dispersion and lack of clear trends in the empirical data themselves.

Discussion—This model does not aim to replicate real-world networks in detail but serves as a simplified synthetic proxy, implementing a two-step mechanism: first forming a backbone structure, then incrementally adding edges to achieve a target degree distribution. This approach offers a new method for generating spatial networks and, from a practical perspective, provides a framework for systematically studying how street network structures impact processes like traffic dynamics, CO₂ emissions, and beyond. However, the model does not capture certain disordered aspects of real-world networks, such as geographic constraints (e.g., rivers, lakes) or orientational disorder. Despite these limitations, the model successfully explains key features of street networks, including the large heterogeneity in BC (and the creation of high-BC paths) and the significant spatial heterogeneity, as reflected in the large spatial Gini coefficient.

This demonstrates that constructing a network in a naive manner—without integrating the MST process—yields structures that fail to account for the large heterogeneity observed in empirical data. Notably, our model does not require any tunable parameters. With a simple input (such as the size of the lattice and fixed node degree fractions f_i based on average values), it offers a minimal and flexible framework for understanding the structure and properties of street networks. As such, it stands as a promising candidate for a foundational, universal model of street networks.

The Python code used herein can be accessed at [45].

Data availability—The data that support the findings of this article are openly available [22].

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